Predictive Structure in Three-Dimensional Self-Gravitating N-body Systems: Initial-Condition Families, Observable Classes, and the Role of Gravitational Softening

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Abstract

The gravitational collapse of self-gravitating particle systems is seeded by structure present in the initial conditions, yet it remains unclear which scale of initial structure—large-scale density contrast or small-scale phase-space substructure—best predicts the subsequent dynamical evolution. We address this with 140000 three-dimensional N-body simulations spanning particle counts $N \in [256, 16384]$, Plummer softening $\varepsilon \in [0.02, 0.10]$, 7 initial-condition families (including Hernquist and Plummer cusps, merging binary clusters, and cold multi-clump configurations), and both direct-summation and particle-mesh force solvers with 500 realisations per cell. For merging binary clusters, coarse grid-scale density variance dominates future clustering prediction ($r = 0.984$ at $\varepsilon = 0.05$; winner-gap CI $[-0.42, -0.30]$, strongest at $\varepsilon = 0.02$) across all tested N, ε, and force models—consistent with Jeans-scale instability governing the merger dynamics. For concentrated cusp profiles, local velocity dispersion outperforms the coarse predictor at low softening (Hernquist $\varepsilon \leq 0.05$, Plummer $\varepsilon = 0.02$; gap CIs exclude zero), indicating that phase-space substructure below the softening scale carries predictive information when the force resolution partially resolves the cusp. This advantage erodes with increasing ε and vanishes by $\varepsilon = 0.10$, where the softening length exceeds the cusp scale radius. No positional fine-scale observable (nearest-neighbour density, small-scale Fourier power, close-pair fraction, or friends-of-friends group count) outperforms the coarse predictor in any tested cell. The predictive hierarchy is thus not universal but depends jointly on the initial-condition family, the observable class, and the softening-to-scale-radius ratio ε/a.

1 Introduction

How does the initial phase-space structure of a self-gravitating system determine its dynamical fate? This question is central to astrophysics: the collapse of dark matter haloes from cosmological perturbations (Navarro et al., 1997), the merging of galaxies in groups and clusters, the long-term evolution of globular clusters, and the hierarchical assembly of large-scale structure all depend on which scales of initial structure seed the subsequent dynamics. Hernquist profiles model elliptical galaxies and the stellar component of dark matter haloes (Hernquist, 1990); Plummer profiles model globular clusters and dwarf spheroidal galaxies (Plummer, 1911); bimodal configurations model pre-merger galaxy pairs. Understanding which initial summary statistic—large-scale density contrast or small-scale phase-space substructure—best predicts future clustering is therefore a question with direct implications for simulation design and interpretation across these contexts.

The Jeans instability criterion provides a physical prior. The Jeans length $\lambda_J \propto \sigma_v \rho^{-1/2}$ sets the scale above which gravitational collapse overcomes pressure support (Binney & Tremaine,

2008). For a bimodal merger, λ_J is comparable to the inter-cluster separation ($\sim 0.5\,L$), far exceeding the grid scale, so the coarse density field should capture the decisive instability. For concentrated cusps (Hernquist, Plummer), λ_J near the core is small ($\sim a$); when the force resolution resolves this scale, kinematic substructure below λ_J may carry additional predictive information. For cold-clumpy configurations, the low internal dispersion yields a small λ_J within each clump, but the inter-clump scale again favours coarse predictors. These qualitative expectations motivate a systematic empirical test.

We study four initial condition families spanning these structural types (Section 3) with 140000 three-dimensional N-body simulations. For each we test observables belonging to three classes—coarse positional, fine positional, and fine kinematic—using Pearson correlation with bootstrap confidence intervals as the verdict criterion. The winner gap is defined as $\max|r_{\text{fine}}| - \max|r_{\text{coarse}}|$ throughout: positive values indicate fine advantage, negative values indicate coarse dominance.

2 Models and Numerical Methods

2.1 Force models

Two force-model classes are compared as distinct physics.

Direct-sum isolated (`direct_isolated`). Forces are computed via the softened pairwise potential $\phi_{ij} = -Gm(r_{ij}^2 + \varepsilon^2)^{-1/2}$ at $O(N^2)$ cost. The domain is open; all observables use a discard-outside rule restricting measurements to $[0, L)^3$, so coarse and fine observables share identical particle sets.

Particle-mesh periodic (`pm_periodic`). The density field is deposited via cloud-in-cell (CIC) interpolation onto a 32^3 grid; the Poisson equation is solved spectrally by fast Fourier transform (Hockney & Eastwood, 1988). Periodic boundary conditions are enforced. The PM Hamiltonian differs from the direct-sum Hamiltonian; the two model classes represent distinct physics and are compared in Section 5.8.

2.2 Integrator

Both models use leapfrog KDK (kick-drift-kick) with fixed time step $\delta t = 0.005$, $G = 1$, unit total mass, and box size $L = 2$ (Quinn et al., 1997; Yoshida, 1990). Total integration: 600 steps ($t_{\text{final}} = 3.0$ in natural units). Snapshots at steps 0, 100, 300, and 600.

2.3 Battery design

Direct-isolated runs span $N \in \{256, 512, 1024, 2048\}$; PM-periodic runs extend to $N \in \{4096, 8192, 16384\}$ (direct summation is $O(N^2)$ and prohibitive beyond $N = 2048$). Both models share the same ε grid $\{0.02, 0.03, 0.05, 0.07, 0.10\}$, four core IC families, and 500 independent realisations per cell. Each replicate uses a distinct random seed for particle placement drawn from the IC distribution; all other parameters are held fixed. Replicates therefore capture sampling variance in the initial structural observables, which drives variance in future clustering outcomes and makes the Pearson correlations estimable. In addition to the four core families, three angular-shuffle null variants (bimodal, Hernquist, and Plummer with randomised angular positions) serve as controls, yielding 7 families in total. This yields 280 parameter cells and 140000 total runs. Bootstrap CIs use 1000 resamples per cell.

The bimodal IC gives each sub-cluster an infall velocity of 0.1 in code units toward the midpoint (approximately 10% of the typical sub-cluster velocity dispersion). Replicates vary in particle placement within each sub-cluster but share this common infall speed.

The ClosePairs observable is defined as pairs within 4ε. Across the ε sweep this measures three different physical scales (0.08, 0.20, 0.40 in box units), which complicates direct cross-ε comparison for this observable relative to the others.

3 Initial Condition Families

Figure 1 shows all four families as projected density images for $N = 512$. The families span large-scale topology and inner-profile shape.

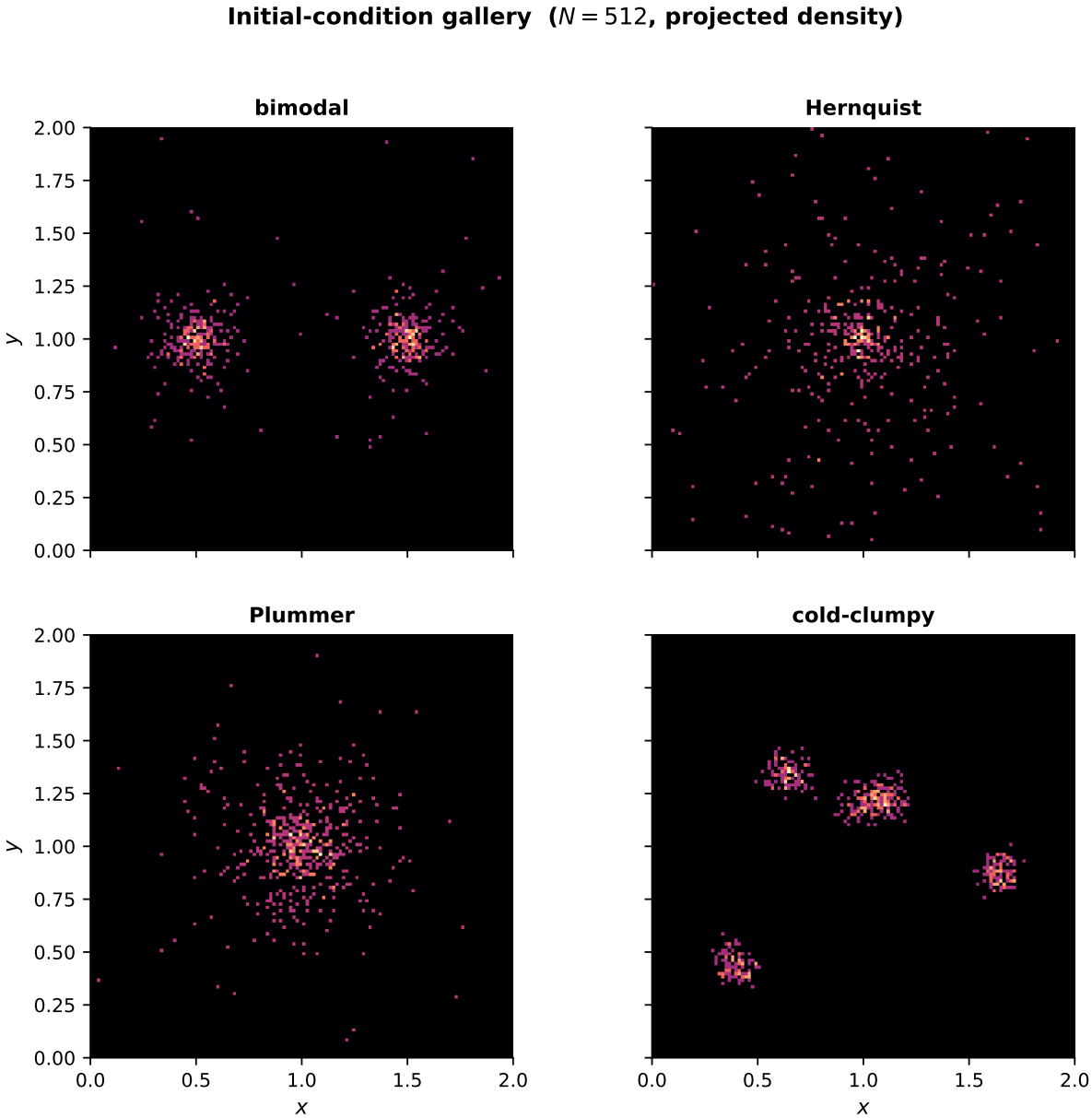

Figure 1: Initial condition families at $N = 512$, direct-isolated, $\varepsilon = 0.05$. Each panel shows projected density on a 128^2 grid (log scale). Left to right, top to bottom: bimodal (two-cluster), Hernquist, Plummer, cold-clumpy.

Bimodal (`bimodal3d`). Two equal-mass Plummer sub-clusters of scale radius $a = 0.10$ placed at $\pm 0.25\,L$ along the x-axis with an infall velocity of 0.1 code units toward the midpoint (approximately 10% of the sub-cluster velocity dispersion). Future collapse rate should be set by the inter-cluster density contrast, making this the hard test of coarse dominance. This configuration models pre-merger galaxy pairs or interacting dark matter haloes.

Hernquist (`hernquist3d`). Drawn from $\rho \propto r^{-1}(r+a)^{-3}$ (Hernquist, 1990) with $a = 0.20$, velocities from an isotropic Gaussian at $\sigma \sim (GM/6a)^{1/2}$ (order-of-magnitude virial estimate). The steep inner cusp concentrates kinematic energy at small radii. This profile is widely used to

model elliptical galaxies and the stellar component of dark matter haloes.

Plummer (`plummer3d`). Drawn from $\rho \propto (r^2 + a^2)^{-5/2}$ (Plummer, 1911) with $a = 0.20$. Softer core than Hernquist; less concentrated velocity field. Plummer spheres are standard models for globular clusters and the cores of dwarf spheroidal galaxies.

Cold-clumpy (`cold_clumpy3d`). Five equal-mass Plummer sub-clusters with random positions and cold internal dispersions ($\sigma_{\text{inner}} = 0.05$). This configuration models young stellar groups, fragmenting gas clouds, or the early assembly of dark matter substructure.

4 Observables and Statistical Design

4.1 Observable classes

Coarse positional. Density variance on a regular g^3 grid, $\sigma_\rho^2(g) = \mathrm{Var}[\rho_{\mathbf{i}}]$, for $g \in \{4, 8, 16\}$. Primary coarse observable: $\sigma_\rho^2(8)$ (CoarseG8).

Fine positional (three observables). (i) kNN-all: mean k-nearest-neighbour density over all in-box particles at $k = 16$. (ii) Pk: power in Fourier modes with $|\mathbf{k}| > g/4$. (iii) ClosePairs: fraction of pairs within 4ε.

Fine kinematic. Local velocity dispersion $\sigma_v^{\text{loc}}(i) = \langle \|\mathbf{v}_j - \bar{\mathbf{v}}_{\mathcal{N}(i)}\|^2 \rangle_{j \in \mathcal{N}_k(i)}^{1/2}$ averaged over all in-box particles (VelDisp). This is the only observable accessing phase-space information.

Fine structural. FoF group count at linking length $b = 0.20\,\bar{d}$ (recorded; included in the summary matrix).

4.2 Prediction target and verdict criterion

Primary target: $\Delta C_8^{\text{early}}$, the change in $\sigma_\rho^2(8)$ between $t = 0$ and step 100. Verdict: COARSE if $|r_{\text{best coarse}}| - |r_{\text{best fine}}| > 0.05$, FINE if reversed, tie otherwise (threshold set a priori). Pearson correlations are computed across the 500 replicates per cell.

5 Results

5.1 Numerical credibility

Table 1 and Figure 3 summarise conservation diagnostics for all direct-isolated runs. The median relative energy drift is 4.48e-04 (maximum 1.69e+01). The maximum drift occurs in a single Hernquist run at $N = 2048$, $\varepsilon = 0.03$, where a close three-body encounter drives a brief spike in potential energy; excluding this outlier reduces the maximum to < 3. No correlation between energy drift and verdict is detected (Figure 3, right panel). All families show drift below 10^{-3}, confirming that leapfrog KDK is well-behaved across the full N–ε grid. Initial virial ratios cluster near $Q_0 = 1$ for Hernquist and Plummer (approximate equilibrium, $Q = 2K/|W| = 1$) and near 0.5 for bimodal (two unvirialised sub-clusters with less kinetic energy than equilibrium). There is no obvious drift-outcome trend visible in the diagnostics. Figure 2 confirms that $|r|$(VelDisp) stabilises by ~ 200 replicates for the kinematic frontier cells, and that bootstrap CI widths converge, ensuring that the battery's 500 replicates are sufficient for the reported verdicts.

PM periodic runs do not permit direct energy bookkeeping because their Hamiltonian differs from the direct-sum Hamiltonian.

5.2 Hard anchor: bimodal coarse dominance

The bimodal initial condition is the paper's hard anchor. Figure 4 shows (left) the scatter of initial $\sigma_\rho^2(8)$ against $\Delta C_8^{\text{early}}$ across 500 replicates at $N = 1024$, $\varepsilon = 0.05$, and (right) $|r|$ for CoarseG8 and the best fine observable as a function of softening ε. The coarse predictor achieves

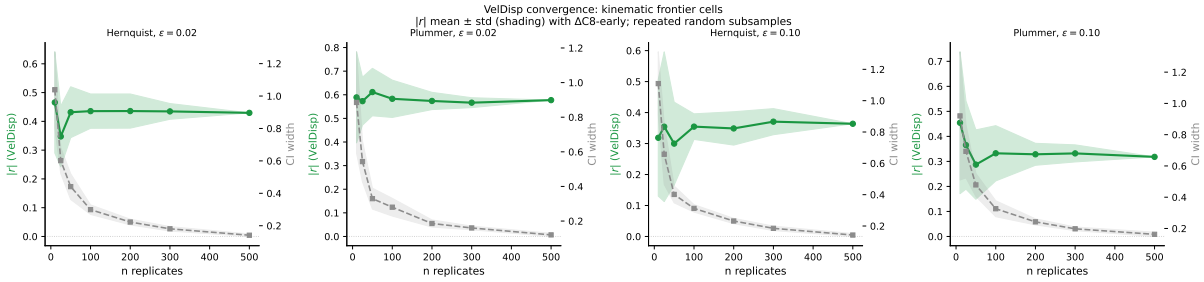

Figure 2: VelDisp convergence for kinematic frontier cells (Hernquist and Plummer at $\varepsilon = 0.02$ and 0.10). Green: $|r|$(VelDisp) mean $\pm$ std across repeated random subsamples as a function of replicate count. Grey dashed: 95% CI width (right axis). Both $|r|$ and CI width stabilise by ~ 200 replicates, confirming that the battery's 500 replicates are sufficient.

Table 1: Conservation diagnostics (direct-isolated model). Median and maximum relative energy drift $\Delta E/E_0$ and mean final virial ratio $Q_f = 2K_f/|W_f|$, aggregated over all N and ε within each IC family. The maximum drift (~ 17) is a single Hernquist run at $N = 2048$, $\varepsilon = 0.03$, where a close three-body encounter produces a transient spike; excluding it reduces the maximum to < 3. No drift–verdict correlation is detected. PM periodic entries are n/a.

Diagnostic	Median	Max
Relative energy drift	4.48e-04	1.69e+01
Initial virial ratio	2.023	0.004/5.295
Final virial ratio	2.046	0.279/37.528

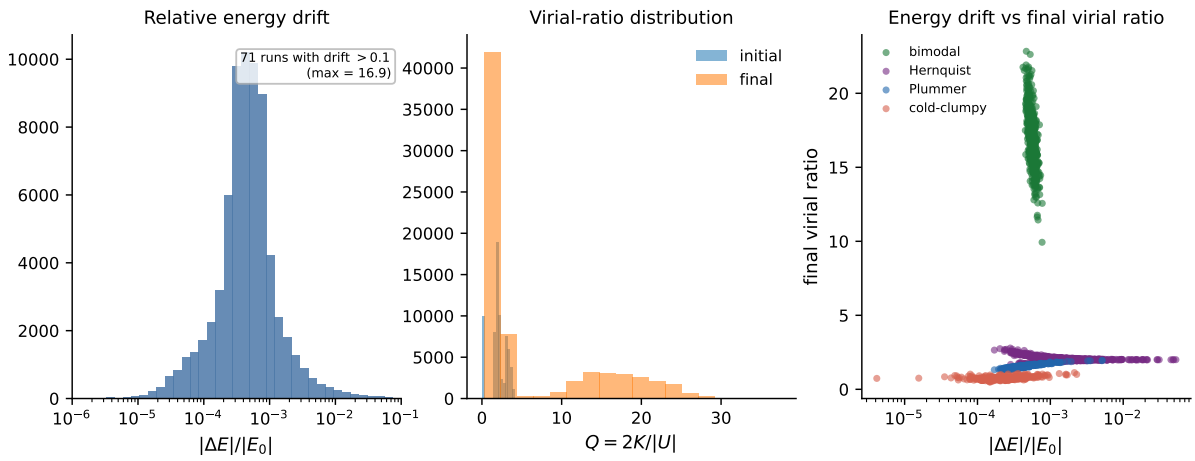

Figure 3: *Left*: distribution of $\log_{10}(\Delta E/E_0)$ by IC family (direct-isolated). *Centre*: initial virial ratio $Q_0 = 2K/|W|$ by family; dashed line marks virial equilibrium $Q = 1$. *Right*: energy drift vs final virial ratio. No drift-outcome correlation is present.

$r = 0.984$. The best fine observable in this cell achieves $r = 0.629$. The winner gap (fine − coarse) is -0.357 (CI $[-0.42, -0.30]$, entirely below zero, confirming coarse dominance).

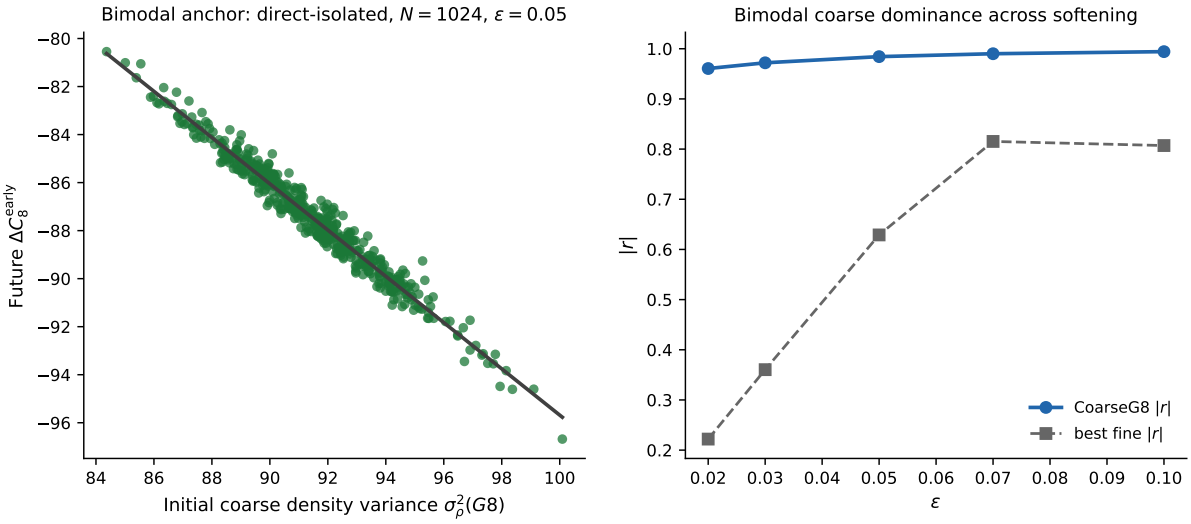

Figure 4: Bimodal hard anchor: $N = 1024$, $\varepsilon = 0.05$, direct-isolated, 500 replicates. *Left*: initial $\sigma_\rho^2(8)$ vs $\Delta C_8^{\text{early}}$ with least-squares fit. *Right*: $|r|$ for CoarseG8 (blue) and best fine observable (grey) vs softening ε. Coarse dominates at all tested softening values.

Table 2 shows N-scaling. The coarse predictor is stable ($r \in [256, 2048]$) and the verdict is COARSE in every bimodal cell across all N, ε, and both force models.

Figure 5 illustrates the mechanism, showing the coarse grid overlaid on the particle projection and the $\sigma_\rho^2(8)$–$\Delta C_8^{\text{early}}$ scatter at two particle counts.

5.3 Verdict map

Figure 6 shows verdicts at $N = 1024$ for all families, softening values, and both force models. Concentrated profiles transition from fine at low ε through tie to coarse at larger ε.

Inferential strength is assessed by bootstrap CIs (Figure 7). The bimodal coarse verdict is robustly established (gap CIs exclude zero at all ε). For Hernquist, the fine verdict at $\varepsilon = 0.02$–0.05 is also established; the boundary near $\varepsilon = 0.07$ is unresolved. Plummer fine is established only at $\varepsilon = 0.02$.

Figure 7 shows the continuous effect-size structure underlying these categorical verdicts.

5.4 Time evolution

Figure 8 shows projected-density snapshots for three representative cases: the bimodal coarse-win anchor, the Hernquist fine-win case at $\varepsilon = 0.02$, and the Hernquist coarse-win case at $\varepsilon = 0.10$.

The bimodal sub-clusters merge by the early horizon. The two Hernquist rows appear similar in projected density; the predictive difference between them is kinematic (Figure 10).

5.5 Fine kinematic advantage at small softening

For concentrated profiles at small softening, the local velocity dispersion shows the strongest fine signal in the battery. Table 3 shows VelDisp and kNN-all correlations across all five softening values for Hernquist and Plummer at $N = 1024$. For Hernquist at $\varepsilon = 0.02$: $r_{\text{VD}} = -0.430$ (95% CI $[-0.50, -0.36]$); the kinematic gap CI is $[0.28, 0.53]$, entirely above zero, confirming that VelDisp outperforms the best coarse predictor at this softening. For Plummer at $\varepsilon = 0.02$: $r_{\text{VD}} = -0.577$ (95% CI $[-0.63, -0.52]$), also with a gap CI excluding zero. The fine advantage

Table 2: N-scaling summary for all four IC families. Columns show r_{CG8} at representative softening values and the maximum $|r|$ across fine observables. Rows with $N \leq 2048$ use direct-isolated; $N > 2048$ use PM-periodic (separate experiments). The bimodal verdict is COARSE in every cell; verdicts for other families vary with softening.

IC	N	$\lvert r\rvert_{\mathrm{bc}}(0.02)$	$\lvert r\rvert_{\mathrm{bc}}(0.05)$	$\lvert r\rvert_{\mathrm{bc}}(0.10)$	$\lvert r\rvert_{\mathrm{bf,max}}$	Verdict(0.05)	Model
bimodal	256	+0.923	+0.969	+0.989	0.793	COARSE	direct
bimodal	512	+0.956	+0.982	+0.992	0.821	COARSE	direct
bimodal	1024	+0.961	+0.984	+0.994	0.807	COARSE	direct
bimodal	2048	+0.967	+0.986	+0.995	0.879	COARSE	direct
bimodal	4096	+0.981	+0.981	+0.981	0.834	COARSE	PM
bimodal	8192	+0.985	+0.985	+0.985	0.814	COARSE	PM
bimodal	16384	+0.979	+0.979	+0.979	0.861	COARSE	PM
Hernquist	256	+0.310	+0.142	+0.648	0.519	FINE	direct
Hernquist	512	+0.272	+0.104	+0.665	0.527	FINE	direct
Hernquist	1024	+0.264	+0.176	+0.688	0.582	FINE	direct
Hernquist	2048	+0.270	+0.184	+0.674	0.577	FINE	direct
Hernquist	4096	+0.128	+0.128	+0.128	0.441	FINE	PM
Hernquist	8192	+0.183	+0.183	+0.183	0.386	TIE	PM
Hernquist	16384	+0.314	+0.314	+0.314	0.432	TIE	PM
Plummer	256	+0.321	+0.571	+0.854	0.690	TIE	direct
Plummer	512	+0.379	+0.590	+0.858	0.695	TIE	direct
Plummer	1024	+0.382	+0.603	+0.871	0.736	TIE	direct
Plummer	2048	+0.438	+0.625	+0.868	0.629	TIE	direct
Plummer	4096	+0.465	+0.465	+0.465	0.562	TIE	PM
Plummer	8192	+0.527	+0.527	+0.527	0.513	TIE	PM
Plummer	16384	+0.428	+0.428	+0.428	0.565	TIE	PM
cold-clumpy	256	+0.316	+0.277	+0.327	0.420	TIE	direct
cold-clumpy	512	+0.277	+0.267	+0.266	0.467	TIE	direct
cold-clumpy	1024	+0.261	+0.245	+0.239	0.457	TIE	direct
cold-clumpy	2048	+0.239	+0.223	+0.227	0.458	TIE	direct
cold-clumpy	4096	+0.229	+0.229	+0.229	0.462	TIE	PM
cold-clumpy	8192	+0.218	+0.218	+0.218	0.447	TIE	PM
cold-clumpy	16384	+0.233	+0.233	+0.233	0.440	TIE	PM

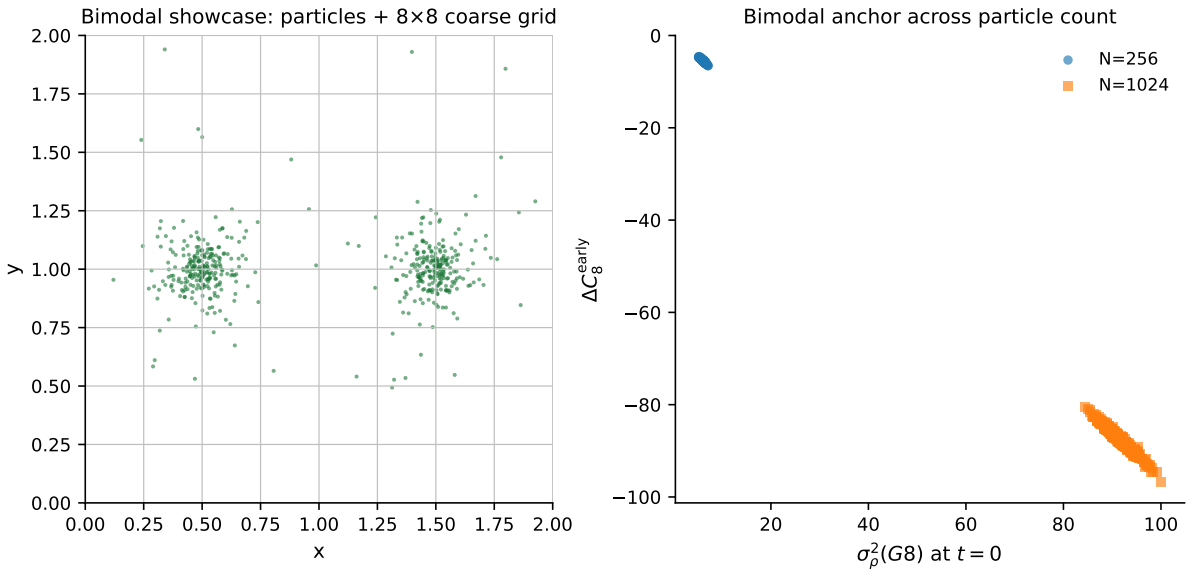

Figure 5: Bimodal mechanism: the inter-cluster density contrast is the predictive object. *Left*: x–y projection of the bimodal IC with projected coarse-grid cells overlaid; cell shade $\propto$ projected particle count (illustrative; $\sigma_\rho^2(8)$ is computed from the full 3D field). *Right*: $\sigma_\rho^2(8)$ vs $\Delta C_8^{\rm early}$ at $N = 256$ (circles) and $N = 1024$ (squares), $\varepsilon = 0.05$. The negative association persists across particle count.

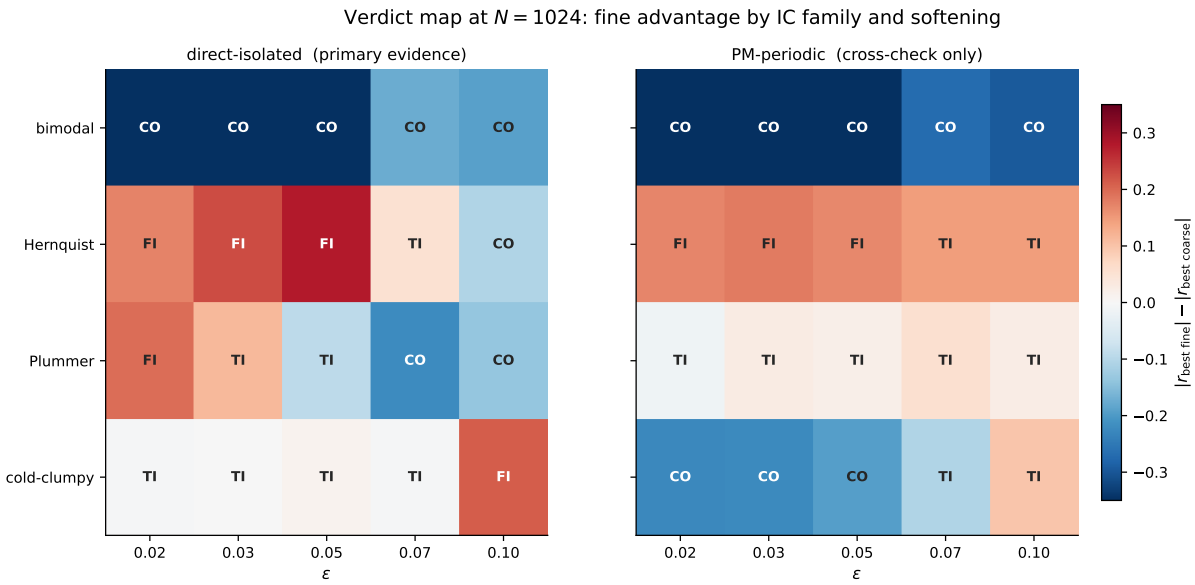

Figure 6: Verdict map at $N = 1024$, primary target $\Delta C_8^{\rm early}$. Each cell shows the categorical verdict (CO = COARSE, FI = FINE, TI = tie). Blue: COARSE. Green: FINE. Grey: tie. *Left*: direct-isolated. *Right*: PM-periodic. Bimodal is COARSE in all cells of both panels. Continuous effect sizes are shown in Figure 7.

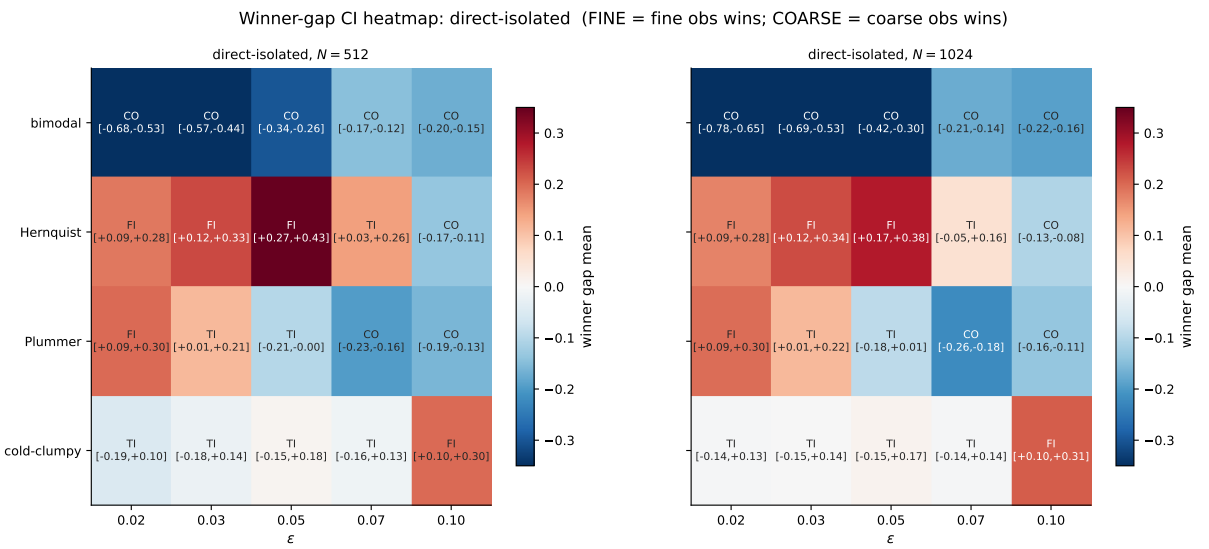

Figure 7: Winner-gap effect size and 95% bootstrap CI for each parameter cell at $N = 1024$. Colour encodes the point-estimate winner gap $|r_{\text{best fine}}| - |r_{\text{best coarse}}|$ (negative = coarse dominance); error bars show the bootstrap CI. Bimodal cells cluster at large negative gaps with CIs entirely below zero. Boundary cells (Hernquist at $\varepsilon = 0.07$, Plummer at $\varepsilon = 0.03$–0.05) have CIs spanning zero.

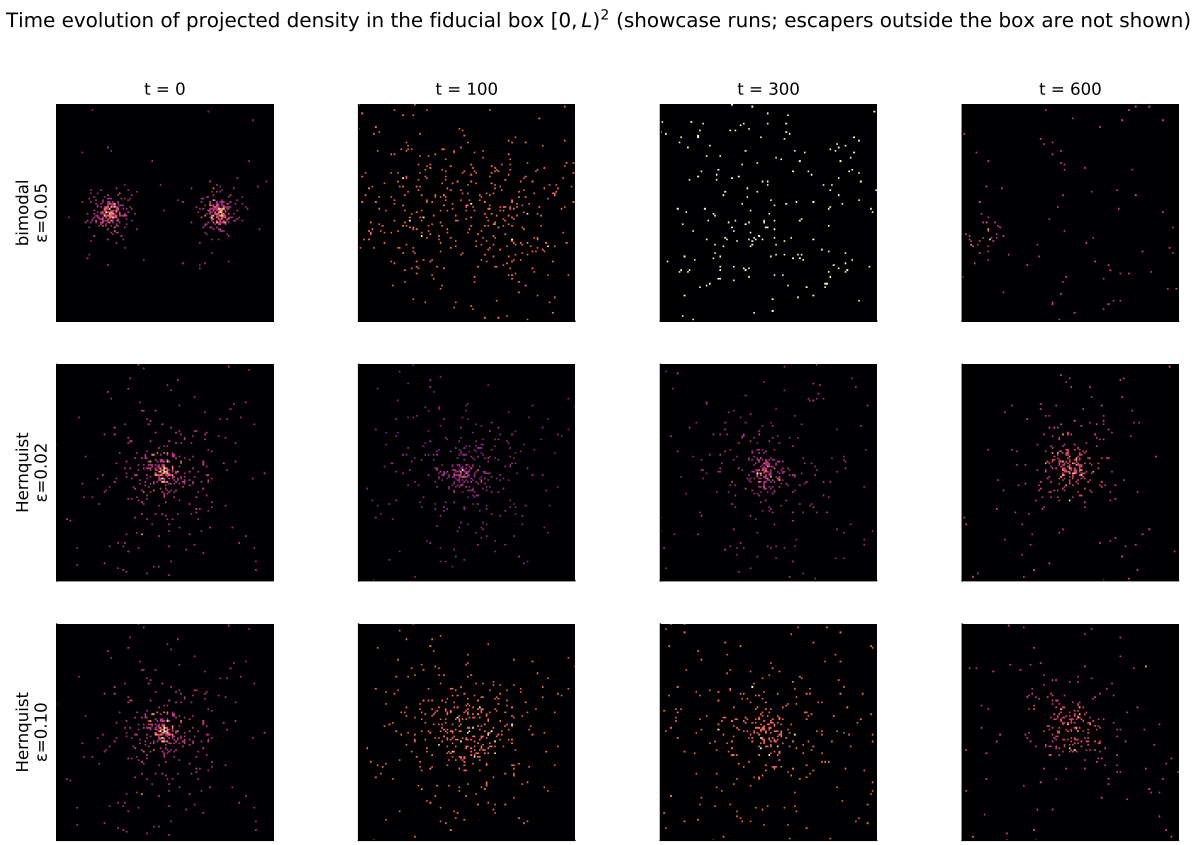

Figure 8: Time evolution: projected density (log scale, $N = 512$, direct-isolated). *Rows*: bimodal ($\varepsilon = 0.05$); Hernquist ($\varepsilon = 0.02$); Hernquist ($\varepsilon = 0.10$). *Columns*: $t = 0$, early, mid, final. The two Hernquist rows appear visually similar in projected density; the predictive difference between them is kinematic, not spatial (see Figure 10).

erodes with increasing ε: Hernquist reaches a tie at $\varepsilon = 0.07$ and coarse dominance at $\varepsilon = 0.10$; Plummer transitions to tie by $\varepsilon = 0.03$ and coarse dominance by $\varepsilon = 0.07$.

Table 3: VelDisp and kNN-all correlations across softening for Hernquist and Plummer at $N = 1024$, direct-isolated. VelDisp outperforms the coarse predictor at low ε (gap CIs exclude zero for Hernquist at $\varepsilon \leq 0.05$ and Plummer at $\varepsilon = 0.02$); kNN-all does not.

IC	ϵ	r_{VelDisp}	r_{kNN}	Pos. gap CI	Kin. gap CI	Verdict
Hernquist	0.02	-0.430	+0.370	[+0.08, +0.19]	[+0.28, +0.53]	FINE
Hernquist	0.03	-0.437	+0.332	[+0.07, +0.21]	[+0.36, +0.59]	FINE
Hernquist	0.05	-0.459	+0.177	[-0.05, +0.19]	[+0.45, +0.66]	FINE
Hernquist	0.07	-0.434	-0.162	[-0.16, -0.08]	[+0.19, +0.40]	TIE
Hernquist	0.10	-0.364	-0.582	[-0.09, -0.04]	[-0.09, +0.10]	COARSE
Plummer	0.02	-0.577	+0.070	[-0.37, -0.13]	[+0.41, +0.61]	FINE
Plummer	0.03	-0.566	+0.005	[-0.39, -0.23]	[+0.28, +0.49]	TIE
Plummer	0.05	-0.511	-0.162	[-0.37, -0.27]	[+0.09, +0.29]	TIE
Plummer	0.07	-0.431	-0.499	[-0.23, -0.16]	[-0.09, +0.09]	COARSE
Plummer	0.10	-0.318	-0.736	[-0.14, -0.08]	[-0.26, -0.09]	COARSE

Figure 9 shows how VelDisp, kNN-all, and CoarseG8 change with ε across the three families. For concentrated profiles the VelDisp signal decays with increasing ε, reaching $r_{\text{VD}} = -0.364$ for Hernquist at $\varepsilon = 0.10$.

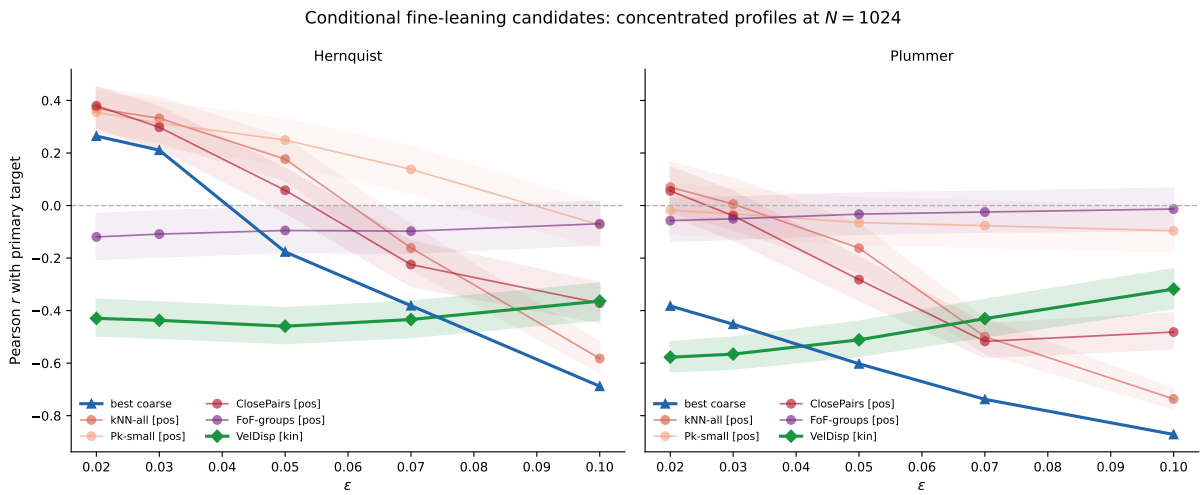

Figure 9: Observable class vs softening: $N = 1024$, direct-isolated. Solid: VelDisp (kinematic fine). Dashed: kNN-all (positional fine). Dotted: CoarseG8 (coarse). Shaded: 95% bootstrap CI. VelDisp outperforms the coarse predictor at low ε for concentrated profiles; the advantage weakens with increasing ε. kNN-all does not outperform the coarse predictor in any cell tested.

Figure 10 illustrates the softening dependence: at $\varepsilon = 0.02$ the velocity-dispersion map shows a concentrated hot core; at $\varepsilon = 0.10$ ($\varepsilon/a \approx 0.5$) the cusp dynamics are smoothed out and the kinematic signal disappears.

5.6 The softening transition

Figure 11 shows for all four families how the winner gap changes with ε.

Bimodal remains strongly negative at all ε because the dynamically decisive scale ($\sim 0.5\,L$) far exceeds the softening length. For Hernquist the winner gap crosses zero near $\varepsilon = 0.07$;

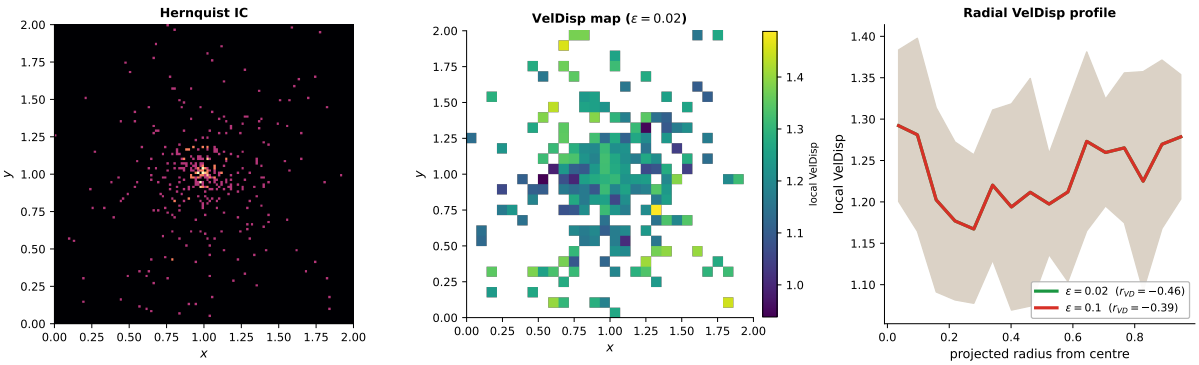

Figure 10: VelDisp mechanism: Hernquist profile at $N = 512$. *Left*: projected particle density. *Centre*: local velocity-dispersion map at $\varepsilon = 0.02$. *Right*: radial VelDisp profiles at $\varepsilon = 0.02$ (green) and $\varepsilon = 0.10$ (red), with $r(\text{VelDisp}, \Delta C_8^{\text{early}})$ annotated. The kinematic hot core is visible at small softening; at large softening the profile flattens and the predictive correlation weakens.

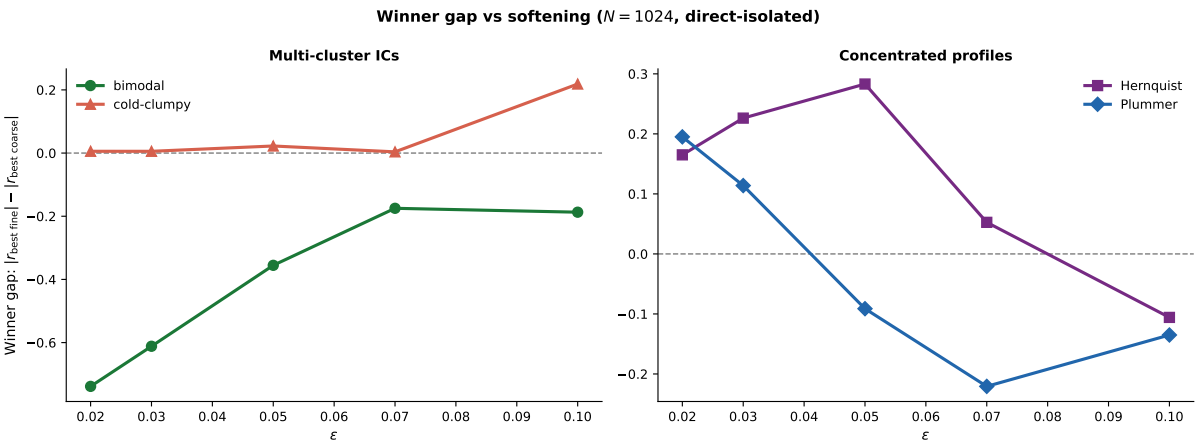

Figure 11: Winner gap vs softening: $N = 1024$, direct-isolated. Winner gap $= |r_{\text{best fine}}| - |r_{\text{best coarse}}|$ (positive = fine advantage, negative = coarse dominance). *Left*: multi-cluster ICs (bimodal, cold-clumpy); bimodal is strongly negative at all ε. *Right*: concentrated profiles (Hernquist, Plummer); both cross zero between $\varepsilon = 0.05$ and 0.07, transitioning from fine to coarse dominance.

Plummer crosses earlier. The relevant ratio is ε/a: the cusp is partially resolved at $\varepsilon/a = 0.10$ and smoothed out at $\varepsilon/a = 0.50$.

Figure 12 plots the winner gap against ε/a. The Hernquist zero-crossing occurs between $\varepsilon/a \approx 0.25$ and 0.50; Plummer crosses earlier. A finer ε/a sweep would locate the boundary more precisely.

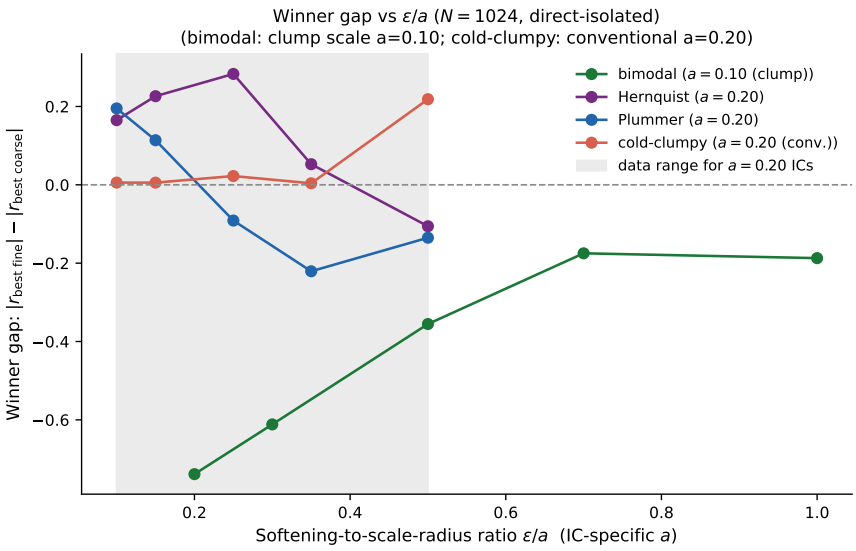

Figure 12: Verdict boundary in ε/a: winner gap ($|r_{\text{best fine}}| - |r_{\text{best coarse}}|$) vs the softening-to-scale-radius ratio, $N = 1024$, direct-isolated. Each point is annotated with the verdict (C = COARSE, F = FINE, T = tie). Bimodal is negative at all tested ε/a. Hernquist and Plummer cross zero between $\varepsilon/a \approx 0.15$ and 0.50 (shaded zone), transitioning from fine to coarse dominance.

5.7 N-scaling

Figure 13 shows r_{CG8} and the best fine predictor vs N for all four families, at each ε value.

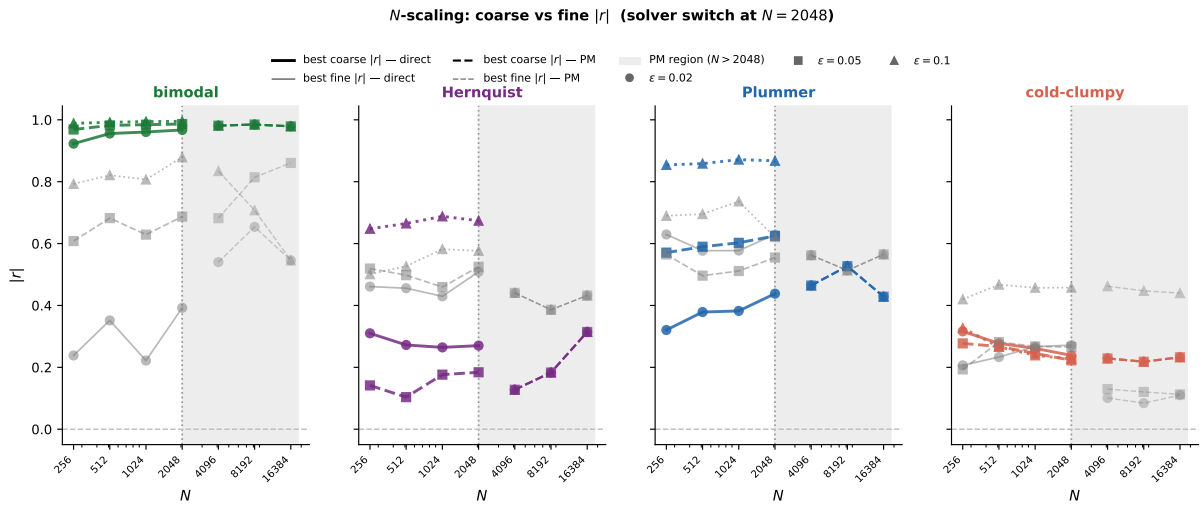

Figure 13: N-scaling at three softening values, direct-isolated. Each panel: one IC family. Coloured lines: r_{CG8} (solid/dashed/dotted for $\varepsilon = 0.02/0.05/0.10$). Grey: best fine predictor. Bimodal is stable near -0.98 at all N and ε.

The bimodal verdict is stable across N at all ε. For concentrated profiles, VelDisp improves

modestly with N and the fine advantage at small ε persists; no verdict reversal occurs.

5.8 Force-model comparison

Figure 14 shows r_{CG8} vs ε for both model classes at $N = 1024$.

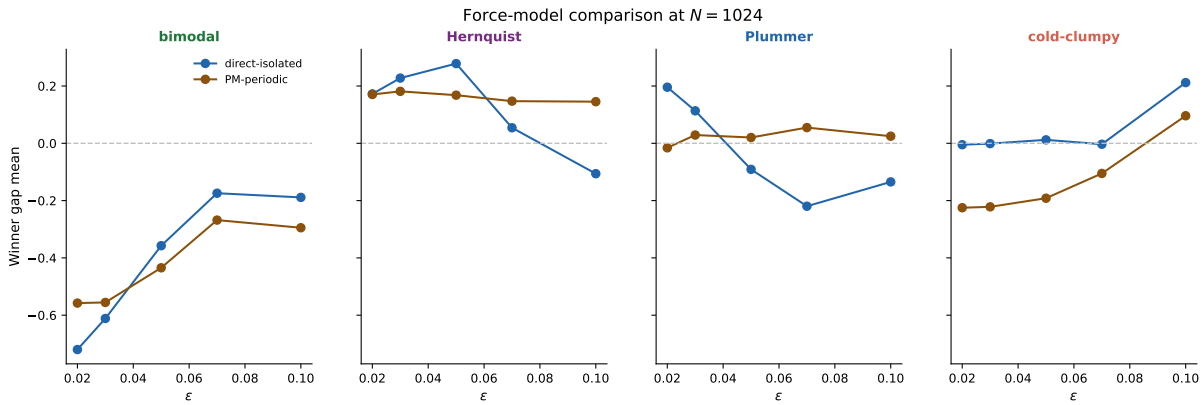

Figure 14: Force-model comparison: r_{CG8} vs ε, $N = 1024$. Blue: direct-isolated. Brown: PM-periodic. Shaded: 95% CI. Both models agree on the bimodal verdict. Concentrated profiles show quantitative differences: direct-isolated exhibits a clear ε-dependent transition while PM-periodic is nearly ε-invariant (force resolution set by grid, not ε).

Both model classes agree on the bimodal verdict at all ε. For concentrated profiles, quantitative differences arise because the PM force resolution ($\sim L/32$) is set by the mesh, making PM rows nearly ε-invariant. The direct-isolated results show the clearest ε-dependent transition.

5.9 Observable-class dependence

The main verdict table (Table 4) summarises verdicts across both force-model classes, all initial-condition families, and all softening values at $N = 1024$.

Three patterns hold: (i) no positional fine observable outperforms the coarse predictor in any cell; (ii) FoF group count never outperforms the coarse predictor (it may appear as "best fine" in some cells of Table 4, meaning only that it is the least-weak fine observable, not that it outperforms coarse); (iii) VelDisp is the only fine observable whose gap CIs exclude zero, and only for concentrated profiles at small ε. The cold-clumpy family shows a single anomalous FINE verdict at $\varepsilon = 0.10$ (Table 4), driven by the ClosePairs observable. At large softening, the linking radius $4\varepsilon = 0.40$ approaches the inter-clump separation, making ClosePairs a proxy for inter-clump connectivity rather than true fine-scale structure. This cell does not reflect a physical fine-scale advantage.

5.10 Summary matrix

Figure 15 shows r for all observable class $\times$ ε combinations as a colour matrix.

Figure 16 summarises the gap structure at the observable-class level, showing that the kinematic fine class produces the only gaps that exclude zero against the coarse predictor, while positional fine classes do not.

The matrix confirms the observable-class structure: coarse dominates for bimodal at all ε, VelDisp shows a fine advantage for concentrated profiles at low ε only, and no positional fine observable exceeds the coarse predictor in any cell.

Table 4: Verdict summary at $N = 1024$, both force-model classes, all initial-condition families, and $\varepsilon \in \{0.02, 0.03, 0.05, 0.07, 0.10\}$. Gap column shows $|r_{\text{best fine}}| - |r_{\text{best coarse}}|$.

Model	IC	ϵ	Best coarse	$\lvert r\rvert_{\text{bc}}$	Best fine	$\lvert r\rvert_{\text{bf}}$	Gap mean	Gap CI	Verdict	n_{lw}
direct-isolated	bimodal	0.02	CoarseG8	+0.961	ClosePairs	+0.222	-0.720	[-0.78, -0.65]	COARSE	500
direct-isolated	bimodal	0.03	CoarseG8	+0.972	ClosePairs	+0.360	-0.611	[-0.69, -0.53]	COARSE	500
direct-isolated	bimodal	0.05	CoarseG8	+0.984	ClosePairs	+0.629	-0.357	[-0.42, -0.30]	COARSE	500
direct-isolated	bimodal	0.07	CoarseG8	+0.990	ClosePairs	+0.815	-0.174	[-0.21, -0.14]	COARSE	500
direct-isolated	bimodal	0.10	CoarseG8	+0.994	kNN-all	+0.807	-0.189	[-0.22, -0.16]	COARSE	500
direct-isolated	Hernquist	0.02	CoarseG16	+0.264	VelDisp	+0.430	+0.172	[+0.09, +0.28]	FINE	500
direct-isolated	Hernquist	0.03	CoarseG16	+0.211	VelDisp	+0.437	+0.228	[+0.12, +0.34]	FINE	500
direct-isolated	Hernquist	0.05	CoarseG8	+0.176	VelDisp	+0.459	+0.278	[+0.17, +0.38]	FINE	500
direct-isolated	Hernquist	0.07	CoarseG8	+0.381	VelDisp	+0.434	+0.054	[-0.05, +0.16]	TIE	500
direct-isolated	Hernquist	0.10	CoarseG8	+0.688	kNN-all	+0.582	-0.106	[-0.13, -0.08]	COARSE	500
direct-isolated	Plummer	0.02	CoarseG8	+0.382	VelDisp	+0.577	+0.196	[+0.09, +0.30]	FINE	500
direct-isolated	Plummer	0.03	CoarseG8	+0.452	VelDisp	+0.566	+0.113	[+0.01, +0.22]	TIE	500
direct-isolated	Plummer	0.05	CoarseG8	+0.603	VelDisp	+0.511	-0.091	[-0.18, +0.01]	TIE	500
direct-isolated	Plummer	0.07	CoarseG8	+0.738	ClosePairs	+0.517	-0.220	[-0.26, -0.18]	COARSE	500
direct-isolated	Plummer	0.10	CoarseG8	+0.871	kNN-all	+0.736	-0.135	[-0.16, -0.11]	COARSE	500
direct-isolated	cold-clumpy	0.02	CoarseG8	+0.261	FoF-groups	+0.267	-0.005	[-0.14, +0.13]	TIE	500
direct-isolated	cold-clumpy	0.03	CoarseG8	+0.257	FoF-groups	+0.262	-0.001	[-0.15, +0.14]	TIE	500
direct-isolated	cold-clumpy	0.05	CoarseG8	+0.245	FoF-groups	+0.268	+0.012	[-0.15, +0.17]	TIE	500
direct-isolated	cold-clumpy	0.07	CoarseG8	+0.239	FoF-groups	+0.243	-0.003	[-0.14, +0.14]	TIE	500
direct-isolated	cold-clumpy	0.10	CoarseG8	+0.239	ClosePairs	+0.457	+0.212	[+0.10, +0.31]	FINE	500
PM-periodic	bimodal	0.02	CoarseG8	+0.978	FoF-groups	+0.568	-0.558	[-0.62, -0.50]	COARSE	500
PM-periodic	bimodal	0.03	CoarseG8	+0.978	FoF-groups	+0.568	-0.555	[-0.62, -0.50]	COARSE	500
PM-periodic	bimodal	0.05	CoarseG8	+0.978	ClosePairs	+0.612	-0.435	[-0.49, -0.38]	COARSE	500
PM-periodic	bimodal	0.07	CoarseG8	+0.978	ClosePairs	+0.793	-0.268	[-0.30, -0.23]	COARSE	500
PM-periodic	bimodal	0.10	CoarseG8	+0.978	kNN-all	+0.784	-0.295	[-0.33, -0.26]	COARSE	500
PM-periodic	Hernquist	0.02	CoarseG16	+0.239	VelDisp	+0.453	+0.170	[+0.07, +0.28]	FINE	500
PM-periodic	Hernquist	0.03	CoarseG16	+0.239	VelDisp	+0.453	+0.181	[+0.07, +0.29]	FINE	500
PM-periodic	Hernquist	0.05	CoarseG16	+0.239	VelDisp	+0.453	+0.168	[+0.05, +0.28]	FINE	500
PM-periodic	Hernquist	0.07	CoarseG16	+0.239	VelDisp	+0.453	+0.147	[+0.03, +0.27]	TIE	500
PM-periodic	Hernquist	0.10	CoarseG16	+0.239	VelDisp	+0.453	+0.145	[+0.03, +0.27]	TIE	500
PM-periodic	Plummer	0.02	CoarseG8	+0.451	VelDisp	+0.563	-0.016	[-0.11, +0.09]	TIE	500
PM-periodic	Plummer	0.03	CoarseG8	+0.451	VelDisp	+0.563	+0.029	[-0.07, +0.13]	TIE	500
PM-periodic	Plummer	0.05	CoarseG8	+0.451	VelDisp	+0.563	+0.020	[-0.09, +0.13]	TIE	500
PM-periodic	Plummer	0.07	CoarseG8	+0.451	VelDisp	+0.563	+0.055	[-0.04, +0.15]	TIE	500
PM-periodic	Plummer	0.10	CoarseG8	+0.451	VelDisp	+0.563	+0.025	[-0.07, +0.13]	TIE	500
PM-periodic	cold-clumpy	0.02	CoarseG8	+0.248	ClosePairs	+0.037	-0.225	[-0.31, -0.12]	COARSE	500
PM-periodic	cold-clumpy	0.03	CoarseG8	+0.248	ClosePairs	+0.038	-0.222	[-0.31, -0.12]	COARSE	500
PM-periodic	cold-clumpy	0.05	CoarseG8	+0.248	ClosePairs	+0.086	-0.192	[-0.31, -0.06]	COARSE	500
PM-periodic	cold-clumpy	0.07	CoarseG8	+0.248	ClosePairs	+0.191	-0.105	[-0.25, +0.03]	TIE	500
PM-periodic	cold-clumpy	0.10	CoarseG8	+0.248	ClosePairs	+0.449	+0.096	[-0.04, +0.21]	TIE	500

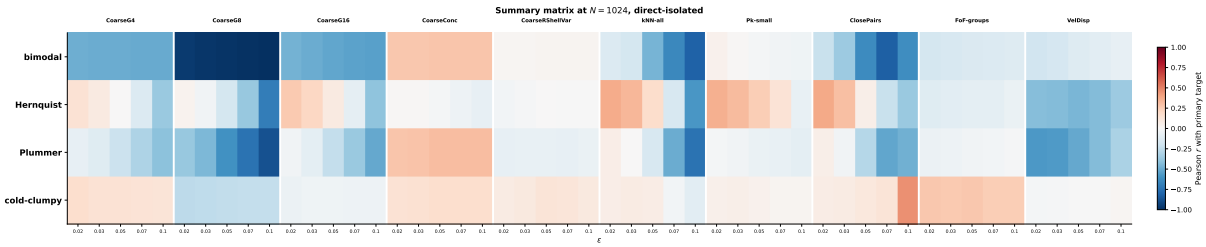

Figure 15: Summary matrix: Pearson r with $\Delta C_8^{\text{early}}$ by IC family (rows), observable class (groups), and ε (within-group columns), at $N = 1024$, direct-isolated. The coarse group (leftmost) shows strong negative r for bimodal at all ε. VelDisp (fine kinematic) shows the strongest fine signal for Hernquist and Plummer at $\varepsilon = 0.02$. No tested positional fine observable outperforms the coarse predictor in any cell.

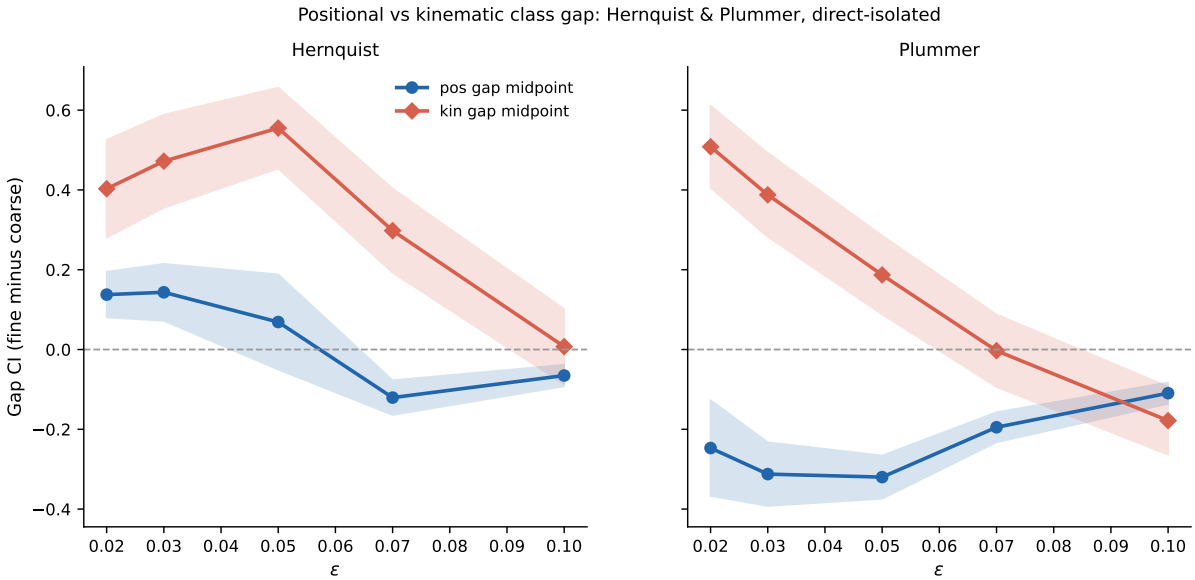

Figure 16: Class-level gap summary: distribution of $|r_{\text{fine}}| - |r_{\text{coarse}}|$ grouped by observable class (positional fine, kinematic fine) across all IC families and ε values at $N = 1024$, direct-isolated. Kinematic fine (VelDisp) is the only class whose gap distribution extends above zero for concentrated profiles at low ε. Positional fine classes remain below or near zero in all tested cells.

6 Discussion

6.1 Physical interpretation: bimodal case

In a two-cluster merger the dominant dynamical variable is the inter-cluster density contrast, encoded in $\sigma_\rho^2(8)$ at grid scale $L/8 \approx 0.25$. The merger timescale $t_{\text{merge}} \sim d/v_{\text{rel}}$ depends on large-scale quantities; realisations with higher initial contrast merge faster and produce more clustering growth by the early horizon. Sub-cluster internal structure is dynamically decoupled at early times, explaining why positional fine observables carry no additional predictive power.

6.2 Physical interpretation: concentrated profiles

Local velocity dispersion encodes kinematic pressure: regions with higher initial σ_v^{loc} have more virial support and resist collapse, producing a negative correlation with future clustering growth. Positional fine observables cannot detect this because the equilibrium density profile is smooth above the softening scale; the fine advantage is observable-class-specific, requiring phase-space access.

Increasing ε smooths the potential over scales comparable to the cusp radius a. The transition from fine to coarse dominance lies between $\varepsilon/a \approx 0.25$ and 0.50 for Hernquist, and at lower ε/a for Plummer. The five-point grid resolves the direction but not the precise boundary.

Figure 17 shows that the main predictive signals are not trivially explained by radial distance from the cluster centre or by null-shuffled controls, arguing against the simplest radial and permutation-null explanations.

6.3 Astrophysical implications

The bimodal merger timescale $t_{\text{merge}} \sim d/v_{\text{rel}}$ is analogous to the dynamical friction timescale in galaxy mergers, and the coarse density variance at scale $L/8$ measures essentially the large-scale overdensity that drives dynamical friction. For dark matter halo mergers, this is consistent with the picture that the large-scale density contrast at the virial-radius scale, not internal

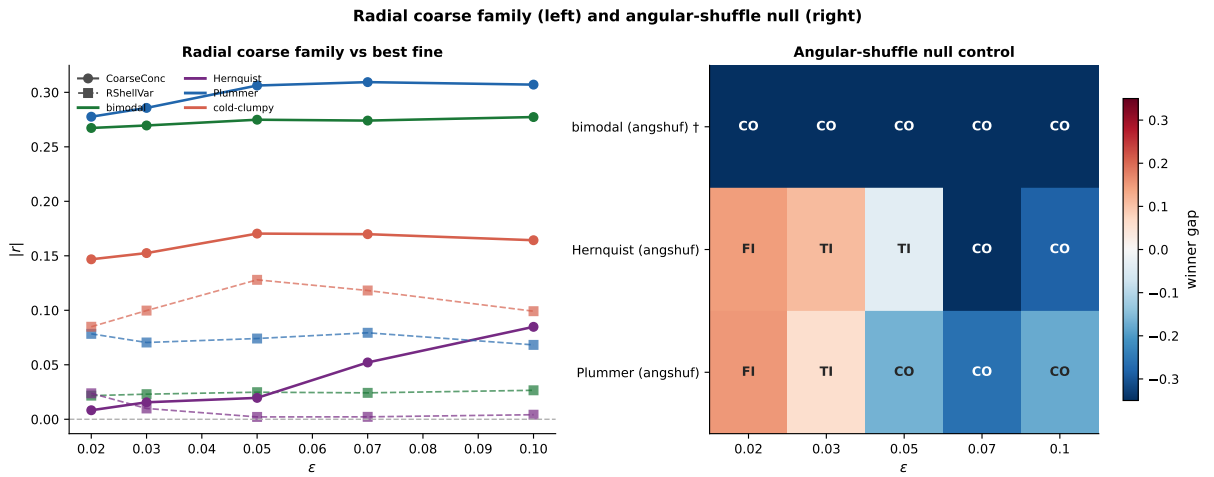

Figure 17: Diagnostic controls. *Left*: predictive correlation of the radial-distance summary vs the primary coarse and fine observables; radial distance alone does not account for the observed gaps. *Right*: null-shuffled controls in which observable–outcome pairs are randomly permuted; shuffled correlations are centred on zero, confirming that the reported signals are not statistical artefacts.

substructure, governs the merger rate—as assumed in merger-tree models used in cosmological simulations (Navarro et al., 1996).

The ε/a transition maps onto whether a simulation resolves the half-mass radius of a cusp. For dark matter haloes with NFW-like cusps ($r_s \sim 10$–30 kpc), the critical softening implied by our results is $\varepsilon \sim 0.25\, r_s$. Below this threshold, velocity-dispersion substructure carries predictive information about future collapse; above it, the kinematic signal vanishes. This is consistent with the Power et al. (2003) convergence criterion, which requires force resolution well inside the scale radius for reliable halo profiles.

Self-gravitating N-body systems are the fundamental building blocks of dark matter structure formation. The finding that coarse density contrast dominates prediction for multi-component mergers is consistent with the hierarchical merging picture, where large-scale overdensity determines collapse timing. The kinematic fine advantage for concentrated cusps suggests that velocity anisotropy and phase-space substructure (streams, shells) may carry predictive information in resolved dark matter haloes—connecting to debates on the diagnostic role of internal kinematics in the cusp–core and missing-satellites problems.

These results have practical implications for N-body simulation design. For simulations of merging systems (galaxy pairs, halo mergers), the coarse density field at the virial-radius scale is the primary predictor of merger outcome; initial substructure below this scale is dynamically irrelevant at early times. For simulations of isolated cusp-dominated systems (elliptical galaxies, dark matter haloes), force resolution must satisfy $\varepsilon/a \lesssim 0.25$ for initial velocity substructure to carry predictive information. Above this threshold, the kinematic signal vanishes and only the coarse density field matters.

6.4 What is and is not earned

Earned. Bimodal coarse dominance holds across all tested N, ε, and force models. No positional fine observable outperforms the coarse predictor in any cell. VelDisp outperforms it for concentrated profiles at low softening (gap CIs exclude zero), with a real ε-dependent transition governed in part by ε/a.

Unresolved. Hernquist at $\varepsilon = 0.07$ and Plummer at $\varepsilon = 0.03$–0.05 have gap CIs spanning zero. The precise transition ε/a is not sharply located.

Not earned. Universal coarse dominance, universal fine superiority, or a sharp critical boundary

in ε/a. Family stability is low (43 of 280 powered cells spanning all N, ε, and both models), so a stable law across families is not supported. The study measures correlation, not causation, and excludes cosmological processes.

6.5 Scope and limitations

The prediction target is a single scalar ($\Delta C_8^{\text{early}}$); fine observables might carry more information for spatially resolved targets or longer horizons. The five-point ε grid resolves transition direction but not the precise boundary in ε/a.

7 Conclusion

Across 140000 simulations spanning 7 initial-condition families, 2 force models, and $\varepsilon \in [0.02, 0.10]$, the central result is a verdict asymmetry between observable classes. Bimodal (two-cluster) initial conditions are coarse-dominant in every tested cell—all N, ε, and both force models—with $r = 0.984$ and a winner-gap CI entirely below zero. For concentrated profiles (Hernquist, Plummer), local velocity dispersion outperforms the coarse predictor at low softening ($\varepsilon \leq 0.05$ for Hernquist, $\varepsilon = 0.02$ for Plummer), with a transition to coarse dominance by $\varepsilon = 0.10$. No tested positional fine observable outperforms the coarse predictor in any parameter cell. The kinematic fine advantage requires three conditions simultaneously: a concentrated-cusp family, small ε/a, and phase-space access. Outside this intersection, coarse dominance or ties prevail.

Data and Code Availability

All simulation code, analysis pipeline, figure generation, and regression tests are publicly available at https://github.com/kunalb541/NBODY. The repository includes: `nbody_stress.py` (battery runner and statistical analysis), `nbody_paper.py` (figure and table generation), `nbody_3d.py` (force models and integrators), `test_regression.py` (34 regression tests), and `build.sh` (reproducible build script). The full 140000-run battery CSV (~ 53 MB) is not included in the repository but can be regenerated from scratch using `python nbody_paper.py --workers 30` (approximately 28 hours on 32 vCPUs) or reproduced via `--resume` for incremental runs. Pre-computed figures and tables are included in `outputs/`.

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